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Author: Alan J. Ramm

Illustrator: W. E. Terry

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*** START OF THE PROJECT GUTENBERG EBOOK TROUBLE
NEAR THE SUN ***

Trouble Near The Sun

By Alan J. Ramm

**Bull and Skip disagreed about the merits
of the Cerebus III as a space ship. But a ship's
mettle—like a man's—is proved in an emergency!**

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The Inner Planet Fleet's spacegoing salvage vessel *Cerebus III* leaped sidewise as though she was trying desperately to escape from some mythical monster of the spaceways. Inside the instrument filled control room the tiny group of ship's officers, gathered together by Captain Stevens' urgent order, felt their feet leave the deck. They reached wildly for any hold available, the lucky ones clinging desperately with strained muscles, the others jumbled in awkward cursing heaps against the bulkheads.

"You idiot!" Captain Stevens shouted. "Next time give us warning!"

The lateral tubes pilot grinned wryly but didn't lift his eyes from the scanner before him. "Sorry, Sir. There was no time. When one of those calcium faculae come boiling up at you like a cannonball, you pick a new spot in the chromosphere for the ship and get there quick—or you don't arrive at all."

Bull Wright, one of the two men who had been strong enough to keep his hold, slowly untwined his huge fingers from a projection and flexed them. He looked across the room and grinned down at the floor where Skip Allen was struggling to his feet. "How do you like good old Sol from that angle?" he drawled sarcastically. "Different from reading about it in a textbook, isn't it?"

The slim built Ensign quickly came to his feet and automatically adjusted the cap on his red head. "Mr. Experience talking, eh? I wonder why Headquarters hasn't discovered that Ensign Wright sees all, knows all, and blabs all?"

"Lay off it, you two," Captain Stevens ordered. "We've got a real problem this time." He paused dramatically, waving a sheet of spacegram paper in the air. "We've been ordered to find the *Regis* and remove her crew and passengers."

"Why that's Fleet Command's new sun cruiser," Skip gasped. "What's happened to her?"

"Headquarters doesn't know exactly," Stevens replied. "They got part of a message saying her propulsion power controls were jammed and her anti-grav and anti-heat equipment was slowly losing effectiveness. They give

her about four hours before she's falling too fast to contact; and about the same time before she gets too hot to maintain life.

"A laminated layer of charged particles must have whipped across her sending beam about then because her message became garbled and finally faded out."

"Rescue," growled Bull. "That's not our kind of job. We're not outfitted for it. If that bunch of stuffed shirts didn't know enough to navigate through the corona and into the chromosphere, they deserve to die. Why should we risk our necks to save them?"

"We're the only ship near enough to stand a chance of reaching them during the next couple hours. But that's not all. Alistar of Cygnus is on board."

"Alistar of Cygnus?" one of the officers questioned. "Who the hell is he?"

"An inspector from Intergalactic Federation Headquarters. Remember that container of electron stripped nuclei found in Federation Headquarters Building last month?"

The men all nodded.

"The Federation figures it came from one of our nuclei dredges in one of Sol's spots. With all the other crazy things that have been happening lately to throw suspicion our way, our system stands in danger of being ejected from the Federation unless we can clear ourselves. You know what that would mean to our trade activities?"

"That's why they've sent Alistar of Cygnus here to make a preliminary investigation. His report will determine further action. Now suppose that something happens to Alistar? They're sure to think that it's part of some plot we're concocting in this corner of the galaxy.

"We've got to find that ship and get him out. Those are our orders. We do it—or die trying."

"Simple problem," sniffed Bull derisively. "All we have to do is locate a tiny thing like that on the sun. That's easy. Only about two and a half

million miles of circumference to cover."

Captain Stevens smiled. "It isn't that bad. The message that was picked up also gave a rough location. They were about 15 degrees North and near a small sunspot about 70 degrees west of the east limb of the sun as seen from Earth at the time of the message. I have plotted the coordinates. We've been on our way there ever since we began talking.

"Are there any further questions, gentlemen? Then take up your assigned duties. You are all dismissed except Ensigns Wright and Allen."

After all the others had filed out, Captain Stevens turned to the two nervously waiting officers. "I warned you two days ago to lay off. I don't mind your picking one another to pieces, but this feud of yours about the space worthiness of the Cerebus III has the whole crew upset."

Skip and Bull looked guiltily at one another. Stevens went on. "Your bantering may have been amusing to the crew for a few days out of Terra. But now they're sensitive about it, and your panning the ship is making the men nervous. Both of you know a nervous crew is nothing to have on a ship operating within the atmosphere of the sun. Even a subconscious suspicion of the worth of the Cerebus III might cause some technician to make a mistake or hesitate long enough to send everyone to his death.

"Just remember, you two," Stevens warned, "this is your first flight as Ensigns. It's strictly a probation period. A word from me can cancel out that diploma you received from Space Academy, Ensign Allen. And wipe that grin from *your* face, Mr. Wright. You may have spent the last ten years working yourself up through the ranks, but your appointment as Ensign is also subject to my report at the end of this tour of duty."

Both men, standing stiffly at attention, muttered, "Yes, sir."

"You both are officers. Now act like you're supposed to. We should sight the Regis in the next hour. I'm assigning you both to the scanning screens. The minute either of you see the Regis, let me know. I'm going to do the rounds and make sure all equipment is ready."

As soon as Stevens left, Bull and Skip went to the tiny cubbyhole called the Vision Room which was located just off Main Control. There they sat back to back in chairs fixed in the middle of the room. This gave each of them a half circle to keep track of. Between them, they had a full view of the ship's

entire surroundings. Sitting there, surrounded by screens, it was easy to imagine that they were afloat in the chromosphere without a ship deck under them.

Several times the forward motion of their ship carried them close to sunspots. These were carefully skirted before the ship entered the penumbra. The dark appearing central area of each spot, the umbra, yawned like monster pits into an area of magnetic forces that ships avoided if possible unless they were nuclei dredges or contact vessels especially constructed to enter the vortex. Twice they ran through loop-like spaces that appeared on their filtered screens. These were caused by prominences of exploding hydrogen shooting tens of thousands of miles into space. Prominences, the tips of which looped back to the sun's surface.

"By the horned three legged elephants of Callisto," muttered Bull as the ship shot through another loop. "I feel like an astronomical ball being shot through loops of an intergalactic croquet game."

As the Cerebus III skittered to the left to avoid another racing faculae, Skip felt his stomach begin to get as unsteady as the ship's deck. He groaned out loud. And for the first time since he sat down, he jerked his strained eyes from the screens only to meet Bull grinning at him.

"What's the matter, Space Cadet?" Bull jibbed. "Don't tell me all that training of yours didn't include a course in how NOT to get space sick?"

Skip swallowed hurriedly before he spoke. "How this garbage collecting tub can stand this buffeting, I can't figure."

"Nothing wrong with her," Bull rejoined. "Just because you thought that you'd get assigned to some fancy interstellar luxury ship when you graduated, you don't have to take out your disappointment on the Cerebus III. Stop making a good ship your alibi for—"

"That's just fancy talk," Skip interrupted. "All you're trying to do is talk yourself into thinking that this is the same as sitting on a stool in some Martian bar."

Glaring at one another across their shoulders, they slowly became aware of Captain Stevens' voice in the Vision Room doorway.

"Forget about the Regis, gentlemen?" His voice held a tempered edge that could have sliced through the million degree temperature of the corona.

Both men jerked to their screens. Off to the side they could see the Regis low in the chromosphere, hanging over the umbra of a small sunspot about ten thousand miles away.

Stevens' voice was bitter. "First you destroy the crew's morale. Now you're negligent in your assigned duty. That should be enough to wash you both out of the Inner Fleet. You're both cosmic debris the Service can do without. Stay out here in the Control Room where I can keep my eye on you. I'll attend to you both later."

Out in the control room Stevens questioned the communications man. "Get a rise out of her yet?"

"It's hopeless, sir. The interference here is too great for contact. This is actually a double spot if you look, sir. That makes communication impossible due to the reversed polarity of the spots."

Skip and Bull, standing wretchedly unwanted and useless to one side of the room, looked at the small screens on the control panel. They could see the Regis balanced precariously over the center of one spot; off in the distance another spot showed clearly—one of the best leader and follower set-ups they had ever seen.

"Getting low, aren't they, sir?" Malcolm, the Second-in-Command, asked Stevens.

"Down to about 500 miles. They must be using the magnetic field there which is perpendicular to the sun's surface to help counteract their own loss of power. They'll be in the reversing layer shortly."

"It's cooler there too," Malcolm observed. "The whirling effect of the gases sets up a low pressure that reduces the temperature to about 7000 degrees instead of the 11,000 outside the spot where we are."

"We'll have to close and hook on as soon as possible," Stevens directed. "Break out the strongest line we have. When we get within a mile, shoot her out until you contact."

"But, sir, we can't get a line out in that if we try."

"We can—and will, Mister. Follow your orders."

"Yes, sir."

The Cerebus III edged in toward the penumbra. In the control room they staggered as the ship was caught up by the madly whirling rim atoms as they sought a passageway into the quiet area over the cooler gases surrounding the spot. Tensely the ship maneuvered its way toward the Regis.

Once they were within range, Bull and Skip could see on the screens the traction line from Cerebus III snake out toward the Regis. Time after time the line just missed. In all that chaos, making contact was worse than threading a needle with a lasso. At last the line struck and held the Regis amidships.

A sigh oozed from every man in the control room. Stevens looked at his watch. "Two hours gone. Now for the tough part."

He called down to the outer lock room through the intership voice tubes. "We've made contact. Have the communicator tube ready to swing out of the lock when I give notice. We are going to start hauling the Regis in toward us now. Be sure that you batten all contact points tight. The men on the Regis haven't any suits to withstand the radiations of the sun. Their only chance is to walk through that tube once you get in contact. A radiation leak down there will kill them all."

Then he called the engine room. "Open up easy."

The deck of the control room began to throb with the power of the huge ionized particle engines. On the screen the traction line began to straighten between the two ships. Its still slack loop twisted like a dying snake between the forces that played over it. Then it tautened.

"It's moving toward us," Skip said aloud. No one answered him. Their eyes were too tightly fixed to the screens.

Suddenly the Cerebus III began to whip right and left. Stevens roughly pushed the pilot from his seat and made some quick moves on the controls. The yawing stopped, but when they looked at the screens, the Regis was once more at a distance, and the traction line was slack.

Stevens looked at the menacing sides of the sunspot. Actually there were no real sides—they were like the sides of a tornado in a mass of air. Here in the interior of the spot, their main problem was to balance the ship against the force of the rising column of gases from the mouth of the spot inside the sun's photosphere, and to adjust their position to the constantly downward drift of the Regis. There was a maximum distance that they could afford to let the Regis drift downward if they wished to save her. Now that they were close to the photosphere, the drag of the sun's 27 G's was greater than it had been out at 10,000 miles. If they managed to pull the Regis close enough for a transfer, it would have to be in the next hour.

"We'll have to take a chance," Stevens said.

Once more he called the engine room. "Throw them in full when I say go," he instructed.

Bull looked at Skip. He smiled and it didn't take words for his thoughts to become clear. *Now you will see some real power from a good ship.*

"Let her go."

The deck leaped to life, reacting to the blows of countless millions of quanta of light as free electrons attached themselves to the stripped nuclei in the discharge chambers of the ship's engines.

The ship began to whip again; but the agile fingers of Captain Stevens brought her swiftly under control.

Skip smiled at Bull. His glance, too, was full of meaning. It said plainly, *this should show you that this ship isn't as good as you think it is!*

The men in the room had to hang on to the sides as the ship, the sun and Captain Stevens fought a duel with one another for control of the Cerebus III and the Regis.

The thin traction line stretched taut once more between them. How much strain could it take? flashed through Skip's mind. Then one torn end seemed to be floating toward the Cerebus III, the other floated toward the Regis.

For a split second the Cerebus III seemed to hang in space. Then everyone was flung violently in all directions as the force of the Cerebus III's engines on full power, plus the reaction of the ship's freedom from the drag of the Regis, shot the salvage vessel into space and plastered the crew against the nearest bulkheads.

Skip came unwillingly back to consciousness. Some of the control room crew were already on their feet. Others were still lying quietly trying to regain their senses before they tried to get to their feet. A few of the men would never move again.

The thrum of the engines had stopped.

Captain Stevens was cursing silently in front of the controls. "Fouled," he spat. "That damn line must have whipped right into the stern discharge tubes and sealed them up so that we have no forward propulsive power. And look out there." He waved his arm at the screens.

The ship had been flung thousands of miles out from its former position. Once more it appeared to be on the outer edge of the chromosphere. The Regis was not in sight. The constantly changing view in the vision panels indicated that the Cerebus III was spinning.

This Captain Stevens corrected while speaking to the men. "Clean up the mess in here. Get the injured to the sick bay; and take care of the dead. We're in the same difficulty as the Regis. Our anti-gray and anti-heat units are working but the indicators show them losing power slowly. We must free those tubes or sooner or later we'll end up in the sun. Let's get down to the outer lock room and see what we can do."

Bull and Skip followed the rest through the big ship. When they got to the lock room, Malcolm, his face white with pain from some injury, was struggling into a solar suit. One of the crew snapped the helmet over his head and handed him a cutting torch. Before any of the new arrivals could say a word, Malcolm was in the lock and on his way outside.

Suddenly one of the lock tenders exclaimed. "My God, he took a suit with a discharged anti-heat unit filler. If his refrigerant cuts off while he's out there

—"

Bull grabbed another suit from its peg and began to draw it on. Captain Stevens grabbed his arm and shouted.

"Hold it, Ensign, you're under arrest. We'll send—"

"There's a man out there," interrupted Bull. "You said this was do or die, didn't you? Let me go."

Stevens tightened his grip. "Listen you big fool—"

Bull shoved hard. The Captain hit the deck and rolled to the nearest bulkhead. No one said a word. There were too many other things to worry about. A fouled discharge system. A man outside about to die unless someone got to him in time.

Bull slammed the inner lock door shut and opened the outer lock. For a moment he drew back. Although he had been spacing for ten years, this was his first trip into the sun's area. In spite of the filters in his viewing plates, the sun looked like a nightmare. Here where the view was unobstructed, the prominences could be seen in their full terror and the boiling jets of flaming gas seemed ready to reach out and pluck him off the side of the Cerebus III. It took all his will power to step outside the lock and look around for the Second-in-Command. Malcolm was almost to the rear of the ship, making his way slowly. Bull moved out after him.

Suddenly Bull's whole front view seemed filled with an explosion. He clung desperately to the ship although he knew that there would be no perceptible effect of such a small explosion in space. He did not have to look to see what had happened; he knew. Malcolm was gone.

At least it had been quick and painless, thought Bull. Malcolm's heat unit had failed. And in such a high temperature the change from a solid to a gas had been so sudden that it was actually an explosion. After swallowing hard several times he slowly began to make his way to the stern. It was up to him to clear the tubes now.

He tried to throw the spectacle of Malcolm's disintegration from his mind; but it kept intruding. He had seen many men die; but none so quickly or so completely. The whole sun was now Malcolm's grave. His very atoms were being torn asunder by the constant process of ionization that was taking place in the sun's atmosphere. Somehow the thought of such complete disembodiment was disturbing to Bull's ideas of immortality.

He was jerked crudely back to reality when the ship came up with unbelievable force to flatten him immovably on its side for a moment. While he gasped for breath under the unexpected pressure, he sought madly for an explanation of his predicament. As the pressure slowly lessened, he realized that the ship must have drifted into one of the many clouds of gas constantly expanding or contracting near the sun due to differences in temperature. This time a rising mass had pressed him against the ship. The lateral jet pilots inside were not compensating for the shift because he was outside and a sudden movement might leave him drifting free.

Once he could get to his feet and proceed, he was extra careful to place his magnetized shoes firmly to the ship's shell. In addition, he set out helper lines to act as auxiliary anchors against any unexpected moves the ship might take. When the possibility of a faculae nearing the ship entered his mind, it took his breath away. If that occurred, he knew that the ship's crew would have to throw in the lateral jets to escape. Unless his equipment was in place at the time of the move, the suddenness of the change in direction would leave him free in the chromosphere until his power ran out; or the faculae the ship had been escaping clasped him and exploded him into another Malcolm.

With his mind trampled with fear, and his eyes fixed firmly to the ship's shell, he was surprised to find himself suddenly within reach of the stern tubes. The long traction line had whipped across the four tubes with a force that had annealed parts of the line to both the inside and outside so that the openings were completely covered. This caused the forces in the firing chambers to neutralize themselves since there was no aperture to permit the egress of force in any direction.

Making sure that all his possible anchorage was in place, he braced himself and began burning out the clogged tubes nearest to him.

Back in the Vision Room the crew anxiously watched bits of metal slough off under Bull's torch.

"That guy's sure got what it takes," someone whispered.

Skip spoke without thinking. "He sure has. I guess he meant what he said about the Cerebus III."

Outside, Bull cleaned out the second tube and reached across without thinking to start cutting on number three. Forgetting that he had pulled all his lines as taut as possible, and that a move in the wrong direction would pull some of them free, he felt his feet break loose and in seconds he was floating twenty feet from the ship with only one line connecting him to safety.

Skip spotted him floating clear, but Captain Stevens shouted. "He's almost free. And he won't dare try to haul himself in by that one line or he might tear it loose, too. He's lucky if some of the eddies around the ship don't do that anyway."

Skip made no comment but started for the outer lock room on the double. By the time the rest had figured what he was up to, he was already in a suit and had shut the lock door behind him. Remembering Bull's trouble on the way along the shell, Skip carefully put out his lines and made as much speed as possible. He didn't dare to look to see if Bull was still out there, or if he had broken free. When the tubes were reached, he looked up and drew a deep breath—Bull was still O.K. Bracing himself carefully, Skip drew in on Bull's line foot by foot. Since it was firmly attached at Bull's suit, there was no danger of it pulling out at that point. At last Bull was along side of Skip.



Once more firmly attached to the ship, Bull slumped against it momentarily. It was impossible to communicate between the suits; and the filters in the helmets didn't permit a look into one another's faces.

When Bull looked up after catching his breath, Skip was already at work on the remaining tubes. Placing himself alongside Skip, he added his torch to the work.

Splinter by splinter, chunk by chunk they burned away the traction line debris. To get the tubes completely free, one of them had to climb into the tubes. Bull did this. The hard work in the confined space of the suits caused sweat to pour from their every pore.

Bull was reaching for one of the last scraps of metal when he noticed that his suit was getting hot. The metal fittings in his hands were becoming too warm to hold. In sudden panic once more he remembered Malcolm. Was his own anti-heat unit becoming depleted? Scuttling out of the tube, he found Skip waiting for him. Looking around, the sight which met his eyes didn't at first register. They could no longer see the sun. *They seemed part of it.*

There could be only one explanation. Their work must have taken longer than it seemed. During that time the ship had drifted downward until it was now well within the vortex of a spot. Since the sides seemed to be pressing all around him, the ship was probably within the photosphere. Unless they escaped at once, they would not get away. He began to work his way back toward the lock; Skip following.

Inside the ship Captain Stevens looked anxiously at his instrument panel. Sweat broke out on his forehead. They couldn't wait any longer. He reached for the firing control.

A gasp came from one of the men in the room. "You can't, sir. Not after what they have done."

Stevens instinctively began to withdraw his hand, then stopped. "I must," he whispered. "It's them or all of us. There is no choice."

His hand plunged down on the firing control.

Outside, Bull had been placing his anchor lines carefully as he went along. He kept slightly behind Skip, making sure that all his lines were in place if Skip's weren't. There was no doubt in his mind as to what Captain Stevens would do when he found that he had to make a choice between the two of them dying or the whole crew. Bull knew that the lines would hold against the thrust of the ship's engines. But would they hold both he and Skip? What is more, would he be able to grab that crazy space cadet that had saved his life, before the ship's momentum tore the kid away and beyond reach?

He had little time to conjecture. His feet felt the ship's shell take life. Snaking out his hands as rapidly as his reflexes allowed, he grabbed Skip around the waist with both arms. Instantly the full force of the ship's new direction and Skip's inertia fought a battle centered on Bull's shoulder joints. Long before his bones slipped from their shoulder sockets, Bull felt

the pain of tensed and torn muscles course down his sides and chest. He heard himself screaming far away in his own helmet. But he held on. And then he lost consciousness.

Later. Much later; Bull woke to find himself lying in a bed in the hospital bay of the ship. His arms were stretched high over his head; held firmly in splints under tension.

"So you've decided to come to, eh?"

Bull turned his head to look at the bandaged figure in the next bed.

"Kind of hard to recognize, I guess."

The voice was familiar. "How come all those fancy bandages, Space Cadet?"

Skip's voice was bitter and self-accusing. "Space Cadet is right. I was outside working on those tubes and never noticed we were getting close to the sun. I must have got a full dose of heat. How dumb can a guy get?"

There was a silence between them. Then Bull spoke. "You saved my life out there, Skip. I couldn't have pulled myself in. That took guts. Thanks."

"What do you call that last act of yours? There isn't another man on this ship that could—or would—have held on with those arms like that."

Both men looked at one another. There would be times when experience and formal education might conflict in the future; but the remarks would never have an edge that mutual admiration wouldn't dull.

"Admiring one another's bravery?"

Both men shifted their eyes to the doorway where Captain Stevens stood. Resentment rose in both of them. Did he have to call them down and rub it in at a time like this?

"Well? What have you got to say for yourselves?"

Bull, realizing that there was no sense in trying to condone what he had done, answered with the first thing that popped into his mind.

"Too bad we didn't reach the Regis, sir."

"But we sure tried," Skip added. "The ship's crew gave everything it had. If we couldn't do it, it couldn't be done."

"Nothing to say for yourselves?" Stevens persisted.

"I guess not, Sir," Skip answered for both of them.

Captain Stevens walked over and stood in the space between their beds. "Both of you seem to have missed an important point; in spite of all your experience—and all your learning."

Bull and Skip were puzzled.

"When that line from the Cerebus III to the Regis let go, what happened to us?" Stevens asked.

"Why, we were flung from here to trouble," Bull exclaimed.

Skip's face screwed up into a frown under his bandages. Then he let out a sigh of relief. "Of course."

"Of course what, book boy?" Bull asked irritably.

"Look," Skip explained. "If you pull a piece of string with a steady tension on both ends, what happens when it breaks? That's right. Objects at *both* ends are flung apart."

"So the Regis was flung out of that spot the same as we were," Bull said in amazement.

"That's right," Captain Stevens agreed. "And she was able to hang on in her new position until another rescue ship reached her. She's safe. Alistar of Cygnus is O.K., too."

"So the Federation and the System are happy," Skip added. "Everything seems to have worked out but us. And we're just plain out."

Captain Stevens looked at them. "Some men," he said, "are like a good ship—like the Cerebus III for instance. They don't show what they're worth until they've had some of the polish and shine rubbed off. Isn't that so, Allen?"

Skip nodded mutely.

"Other men," Stevens continued, "are afraid that someone might know more than they do. Don't you agree, Wright?"

Bull's eyes were taking on a new light.

"What I really came here for, Ensigns, was to find out when you two figure that you'll be able to pick up your duties as regular Officers of the Cerebus III? I like to keep good men like you with me."

Bull and Skip looked at one another and grinned.

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THE SUN ***

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